

Quiz 4 — Walking on a Graph

M 375T Mathematical Foundations of Machine Learning — Fall 2026

15 minutes · closed book · no devices · same skills as Practice 5

Name: _____ UT EID: _____

1. The code builds the weight matrix of an undirected graph on the vertices 0, 1, 2, 3. The walker moves with the rule $P_{ij} = W_{ij}/d_i$. [5 pts]

```
import numpy as np
W = np.array([[0, 2, 0, 1],
              [2, 0, 1, 0],
              [0, 1, 0, 1],
              [1, 0, 1, 0]])
```

- (a) Draw the graph: the four vertices, every edge, and each edge's weight.
- (b) Write the degree vector d and the transition matrix P .
- (c) The walker starts at vertex 0, so $\mu_0 = (1, 0, 0, 0)^\top$. Give μ_1 , the law after one step.

Answer:

2. Same graph.

[5 pts]

- (a) Write down, *from memory*, the equation that an invariant measure π of the walk satisfies. (No credit for the answer to (b) alone — the equation is the point.)
- (b) Find π for this graph. It must be a probability vector.
- (c) Verify your equation at vertex 0 by explicit computation.
- (d) In one sentence: what does π describe about a long walk?

Answer:

Total: 10 points. Lowest two quiz scores are dropped at the end of the semester.